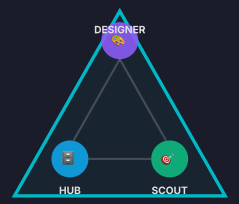
Canadian Tire Corporation
Triangle Rewards · Real-Time Activation

TriStar

Triangle Smart Targeting & Real-Time Activation

AI-powered offer engine that senses context, scores the moment, and delivers
the perfect loyalty offer — before the customer even asks.

Team Details & Artefact links



Triangle Smart Targeting & Real-Time Activation

TriStar

EPAM HACKATHON 2026 · LOYALTY AI PLATFORM

TEAM

Representative: *Ram Naresh, Kartikey Puri*

TEAM MEMBERS

RN **Ram Naresh** — AI Architecture & Backend — Rep

KP **Kartikey Puri** — Full Stack & Integration — Rep

MS **Mohit Soni** — Frontend & UX Design

RS **Riyaj Shaikh** — Backend & DevOps

🎨 **Designer — AI Offer Copilot**

🗄️ **Hub — Lifecycle State**

🎯 **Scout — Real-Time Activation**

What we built: AI-powered three-layer loyalty engine — **Designer** turns plain-English objectives into fraud-validated OfferBriefs; **Hub** tracks lifecycle in Redis; **Scout** activates at the perfect moment using GPS, weather & behaviour signals. Stack: **React 19 · FastAPI · Claude API · Redis · Azure**.

git.garage.epam.com/kartikey_puri/tristar

EPAM Garage · develop branch · Full source code

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Problem Statement

The current loyalty ecosystem operates reactively — members accumulate points but receive limited, generic engagement:

73%	3–5h	0%
offers go unnoticed	manual offer creation	real-time context used

- High-value lapsed members remain completely unengaged
- Offers ignore location, weather & purchase behaviour
- No feedback loop → repeated campaign inefficiencies
- Fraud & over-discounting go undetected before launch
- Siloed tools: no lifecycle visibility draft-to-expiry

CATEGORY

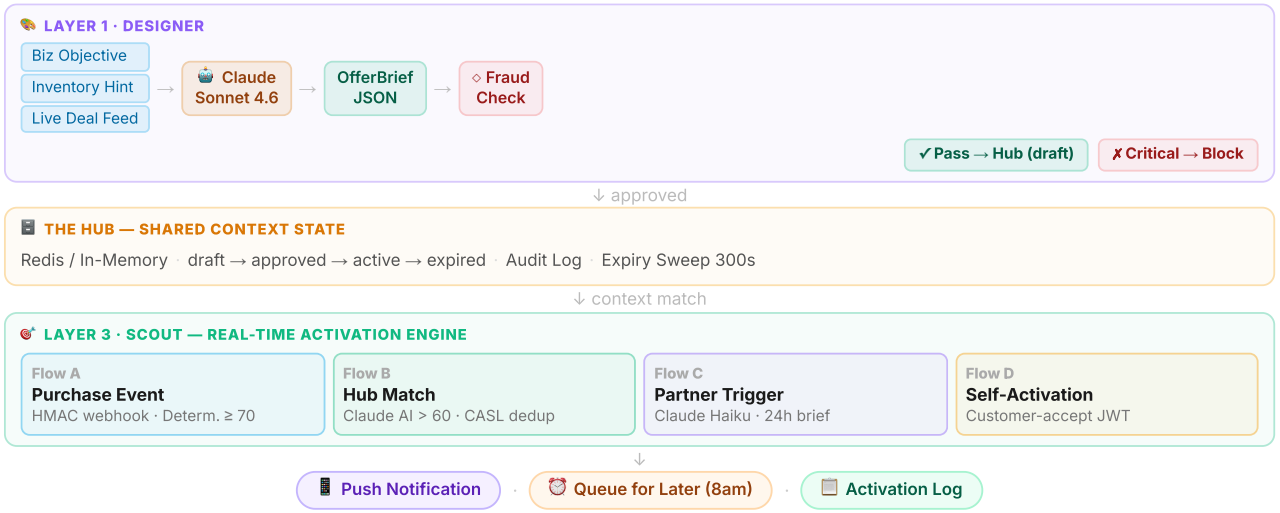
AI-Powered Loyalty Personalisation

Aligned with Customer Experience & Digital Innovation

Solution

TriStar transforms loyalty into an **AI-powered, real-time engagement engine** — delivering the **right offer** to the **right member** at the **right moment**.

SYSTEM OVERVIEW — END-TO-END FLOW



Tech Stack

Next.js 15 · FastAPI · Claude Sonnet 4.6 · Redis · Azure · Pydantic v2

Impact

+25%	80%	<200ms
lapsed member reactivation	faster campaign creation	real-time activation

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Benefits

For Marketers

- Generate validated OfferBriefs in seconds — not hours
- Built-in fraud detection catches over-discounting before launch
- Inventory-driven recommendations increase campaign ROI
- Real-time risk flags surfaced before any offer goes live

For Members

- Offers arrive only when GPS, time & weather align
- Fewer, more relevant notifications — less fatigue
- Seamless, personalised loyalty experience at scale

For the Business

- Full lifecycle visibility: draft → approved → active → expired
- Immutable audit trail — OWASP & compliance ready
- Horizontal scale on Azure · millions of members

Future Enhancements

PRODUCT ROADMAP

Phase 1 — MVP ✓
Now · Hackathon

- Designer + Hub + Scout core
- Claude API brief generation
- 4-pattern fraud detection
- Real-time context scoring
- JWT auth + Redis + Azure

Phase 2 — Intelligence
Q3 2026

- ML-based segment targeting
- A/B testing framework
- Bilingual AI-generated offers
- Live analytics dashboard

Phase 3 — Scale & Reach
Q1 2027

- Partner ecosystem open APIs
- Event-driven low-latency arch
- Mobile SDK + geofencing
- Multi-agent autonomous campaigns

Built-in Guardrails (Phase 1)

🔒 1 offer / member / hr | 🔒 No duplicates / 24h | 🔒 Quiet hrs 10 pm–8 am | 🔒 Critical fraud → hard block

Security: OWASP Top 10 mitigations built in — parameterised queries, Zod + Pydantic input validation, JWT 1h expiry, Azure Key Vault for secrets, PII-free logs (member_id only).

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